

■■■■■ electrical impedance tomography: EIT and ■■■■■

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Open Access Accepted Manuscript for Physiological
Measurement

PRESENTED BY: CHRIS ROCHELEAU

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In-water electrical impedance tomography: EIT and the sea

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COOL! WHY?

- Seawater is very heavy!
- “During diving, increasing pressure causes alveolar compression and a depth-dependent pulmonary shunt that eventually results in alveolar collapse and cessation of gas exchanged”
- EIT systems used to monitor gas exchange in cardiopulmonary systems can better help us understand breathing in aquatic mammals, and protect the safety of SCUBA divers
- However, we need to address the fact seawater has conductivity approximately an order of magnitude larger than the body...

MODELING CURRENT FLOW

- Used a FEM to assess current flow from one electrode to the next
- B shows current flow of body in air
- C-F show current flow of body in seawater, with variable snap sizes covering the electrode
- In each case, once the current has extended beyond the snap, it will traverse outside the body

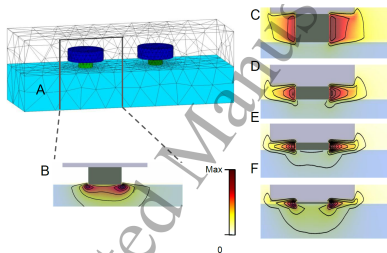


Figure 1: Current flow between two body-surface electrodes in a model in air and saltwater. A) FEM of a body (light blue) with two electrodes (green) covered by a non-conductive snap (dark blue). B-F) Current density in a central slice of the FEM, shown with respect to the colourbar and contour levels as iso-current lines. B) Current into the body when the outside is non-conductive air. Images C, D, E and F show current in seawater with conductivity 5x the body. The thickness of the exposed electrode varies from 0.2 (C), 0.1 (D), 0.05 (E), 0.02 (F).

COVERING THE BODY

- If current will escape the body when only electrodes are covered, we must cover the entire row of electrodes
- Created FEM in EIDORS of body with varying thickness of sea water around the boundary
- Compared voltage difference between object with and without contrasting target in body using skip 0 current pattern
- As expected, signal decreases rapidly with increasing thickness. But signal remains if a small enough water layer can be maintained

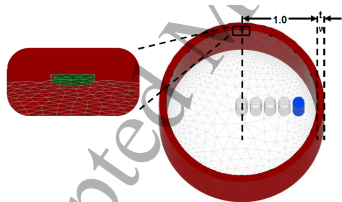


Figure 2: Finite element model used, sliced at the electrode plane. Right: a cylindrical body is simulated with radius=1.0 and surrounded by 16 electrodes (green) and a conductive seawater layer of thickness t_w . Left: Inset of hollow electrode (green). Cylindrical targets are placed at radial distances from the centre.

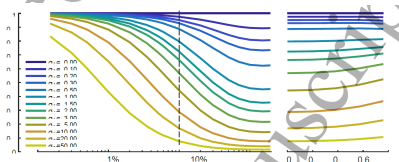


Figure 3: Difference signal, $\|V_t / V_0 - 1\|$ with skip $S=0$, as a function of (left) water-layer conductivity (σ_w) vs medium conductivity, $\sigma=1$) and thickness (t_w), for a target placed at the centre; and (right) as a function of target radial position (for $t_w=10\%$, corresponding to the dotted line). Signals are normalized with respect to an air ($\sigma_w=0$) outer layer.

RECONSTRUCTION FEASIBILITY

- Using GREIT algorithm reconstructions of target at $r = 0.75$ and $\sigma_w = 10$ were used to test sensitivity to skip patterns and water thickness
- “conductive surroundings ‘push’ the image toward the centre, and this effect increases with t_w ”
- “Additionally, the presence of a seawater layer dramatically reduces the image amplitude”
- “If $t_w = 2\%$ and $\sigma_w/\sigma < 20$ then the signal strength is above 50%, the image amplitude is 85% and the image distortion is relatively small.”

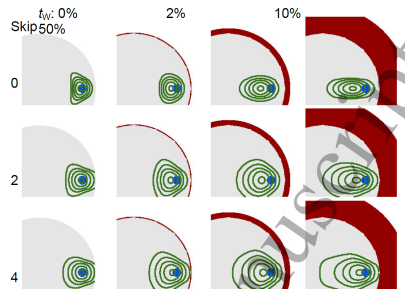


Figure 4: Reconstructions as a function of skip pattern (0,2,4 from left to right) and the thickness of the conductive water layer ($\sigma_w=10$), shown in red (t_w , as a % increase in medium radius, where 0% indicates no water layer). A small cylindrical target (blue cylinder) is simulated, and the contour lines show the reconstructed image as a fraction of maximum (20%,40%,60%,80%,95%).

HARDWARE

- Developed a belt to maintain less than 2mm of water between body and insulating layer
 - 16 Conductive carbon film electrodes
 - Outer neoprene layer to contain electrodes, wires, and connectors.
- Data collected using Sentec pioneer set



Figure 5: Modified EIT belt and its use on a dolphin. Left: The EIT electrodes are made of conductive carbon film with a rear snap connector to the cables. All other parts are neoprene, including an outer layer to prevent current leakage. Right: The belt placed on a dolphin during habituation.

HUMAN TEST

- Two belts of 16 electrodes
- Skip 4 Current Pattern
- Subject supine at 1m depth, breathing through snorkel
- Reference frame at end inspiration

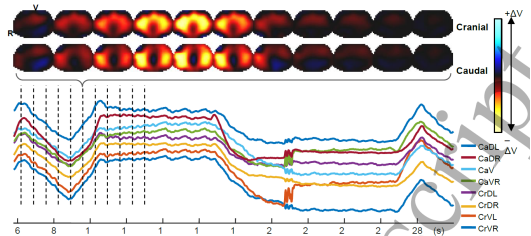


Figure 6: EIT results from an in-water healthy human volunteer. Top row, transverse EIT images at the Cranial and Caudal EIT belt positions (with orientation (V)entral and (R)ight as shown). Right: colour mapping for increases and decreases in air volume. Images correspond to the dotted lines in the EIT waveforms below. Regions of interest are shown for (Cr)anial/(Ca)udal, (V)entral/(D)orsal, and (R)ight/(L)eft. Note the electrode disconnect artefact at $t=21$ s. The breath between $t=10$ s and $t=27$ s represents an inspiratory breath hold for 7 seconds, followed by an expiratory breath hold for 6 seconds.

DOLPHIN TEST

- Subject is a dolphin, reconstructions rotated such that dorsal (posterior) direction is up
- Single row of 16 electrodes
- Skip 0 current pattern
- EIT volume signal compared to spirometer signal
- Regional volumes and reconstructions show a delay in expiration of dorsal lungs, may be a reflection of pneumonia

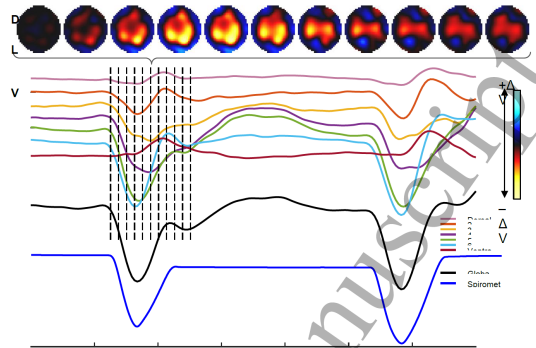


Figure 7: EIT results from an in-water dolphin with suspected bronchitis (Dorsal, D, Left, L, and Ventral V, indicated). Time (bottom) is seconds since the start of recording. Two breaths (exhale then inhale) are shown. EIT waveforms correspond to regions of interest from the Dorsal (top) to Ventral (bottom) of the thorax cross section. The Global EIT waveform is shown in black, and spirometer volume in blue. Lung volume corresponds to an increase in waveform. EIT images (bottom) correspond in time to the dotted lines. The EIT reference is the average during $221.5 < t < 222$ s. Note the initial air exhale from the ventral lung, followed by the central regions.

CONCLUSIONS

- “This paper illustrates the feasibility of in-water EIT; however many challenges remain”
 - Tight neoprene vest may restrict lung capacity and reduce lung compliance
 - Movement artefacts may appear and cause electrode disconnection
 - “We consider the management of electrode contact quality as the most important unsolved problem for EIT measurements in water”
 - “The use of wired EIT cables is additionally cumbersome in water, especially for unrestrained animals”
- “EIT measurements are thus feasible in marine environments, and can be used to explore diving physiology and lung disease in diving mammals.”